Mohamed Bahi YAHIAOUI

Deep Learning Research Engineer

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AI model safety and reliability, with applications in physics.

PROFESSIONAL EXPERIENCE

CEA Cadarache — Research Engineer · Nov. 2022 - Nov. 2025

- Developed an out-of-distribution (OOD) detection method achieving state-of-the-art performance.
- Built dynamic neural network surrogate models for fission gas transport and accumulation over the full lifetime of a reactor, achieving a 100× speed-up over reference simulations.
- Developed neural network surrogate models for a Gibbs energy minimizer.

CEA Saclay — Research Engineer | AAIMME Project Feb. 2022 - Sept. 2022

- Worked on inverse reconstruction with deep learning for PET imaging: estimation of gamma photon interaction position and time from complex detector signals.
- Developed uncertainty-aware approaches providing both point estimates and associated uncertainty on interaction parameters.

CEA Saclay — Research Intern Apr. 2021 - Sept. 2021

- Developed a multi-task deep learning model for identification and intensity estimation of radioactive sources, with explicit uncertainty quantification.
- Studied Density Neural Networks, Bayesian Neural Networks, and information-theoretic measures for uncertainty estimation.

EDUCATION

Université Grenoble Alpes

PhD in Computer Science (Defense: November 27, 2025)

Topic: “Neural network surrogate models with a novel monitoring approach to accelerate multiphysics simulations”

École des Mines de Saint-Étienne

Master of Science (MSc) in Applied Mathematics

Coursework: stochastic modeling, statistical learning, numerical modeling.

Institut National des Postes et Télécommunications, Rabat

Engineering Degree (MEng) in Data Science

Coursework: machine learning, statistical modeling, software engineering, big data, distributed systems.

SKILLS

Scientific machine learning: Physics-informed neural networks (PINNs), neural surrogate modeling, operator learning

Core ML: Deep learning, representation learning, computer vision, multi-task learning

Reliable AI: Uncertainty quantification, out-of-distribution detection, calibration, robustness evaluation

Modeling & computation: Inverse problems, numerical modeling, optimization, statistical learning

Software & engineering: Python, C++, Linux, Git, reproducible experimentation, profiling, deployment

Languages: French (bilingual), English (bilingual), Arabic (native)